

The Second-Order Correlation Measure of Zadoff-Chu Sequences

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Abstract For an integer $r \geq 1$, the r th-order correlation measure (or the correlation measure of order r) of a sequence was introduced in cryptography as a measure of immunity against correlation attacks. When $r = 2$, it can be viewed as the maximum value in a subset of magnitudes of partial-period autocorrelation for a given sequence. In this paper, we determine the second-order correlation measure of Zadoff-Chu sequences of any period and show that it is invariant to the design parameter for them.

key words: Correlation measure, Zadoff-Chu sequence, partial-period autocorrelation.

1. Introduction

During the past several decades, numerous research on the pseudo-randomness of sequences has been done in the fields of sequences, cryptography, and communications [1], [2]. Although the required criteria for pseudo-randomness depend on various circumstances under which the sequences are employed, at the core is how easily a sequence is distinguishable from its cyclically shifted copies and other sequences. One important measure of pseudo-randomness is the *correlation measure of order r* [3] or the *r th-order correlation measure* for an integer $r \geq 1$, which is primarily inspired by cryptography. It is a measure of immunity against correlation attacks.

Let $\{a(n)\}$ be a binary sequence of length N . For a positive integer r , the r th-order correlation measure or the correlation measure of order r of $\{a(n)\}$, denoted by $C_r(a)$, is defined as

$$C_r(a) = \max_{\substack{\{i_1, i_2, \dots, i_r\}, \\ 1 \leq M \leq N - i_r}} \left| \sum_{n=0}^{M-1} a(n + i_1) a(n + i_2) \cdots a(n + i_r) \right|,$$

where $M \in \mathbb{Z}$, and $\{i_1, i_2, \dots, i_r\}$ is an integer-valued set with $0 \leq i_1 < i_2 < \dots < i_r < N$. From the cryptographic point of view, it is desirable that $C_r(a)$ should be as small as possible in order that $\{a(n)\}$ as a stream cipher has high immunity against correlation attacks [4]. To researchers in the field of sequences, the second-order correlation measure $C_2(a)$ may be particularly of interest since it is closely related to the autocorrelation of $\{a(n)\}$. In fact, it was investigated in [3], [5]-[8]. In particular, Alon *et al.* [5], [7] showed that for an arbitrary binary sequence $\{a(n)\}$ of length N ,

$$\sqrt{\frac{N}{2} \ln \frac{N(N-1)}{2}} \leq C_2(a) \leq \frac{7}{4} \sqrt{N \ln \frac{N(N-1)}{2}}$$

if N is sufficiently large.

In this paper, we extend the concept of the second-order correlation measure to nonbinary sequences, and relate it to their partial-period autocorrelation. We then focus on the second-order correlation measure of Zadoff-Chu

sequences. By exploiting the known results on the partial-period autocorrelation of Zadoff-Chu sequences, we compute the second-order correlation measure of Zadoff-Chu sequences of any period, and show that it is invariant to the design parameter for Zadoff-Chu sequences of a given period. These results may act as an upper bound on the second-order correlation measure of a best polyphase sequence with perfect periodic autocorrelation.

The outline of the paper is as follows. In Section II, we give some preliminaries as well as the link between the second-order correlation measure and the partial-period autocorrelation of Zadoff-Chu sequences, we determine its second-order correlation measure in a closed form in Section III. Finally, we give some concluding remarks in Section IV.

2. Preliminaries

Throughout the paper, we denote by $\langle x \rangle_y$ the least nonnegative residue of x modulo y for an integer x and a positive integer y , and denote by $\mathbb{Z}_N$ the set of nonnegative residues modulo N for a positive integer N . Let $\{a(n)\}_{n=0}^{N-1}$ be a q -ary polyphase sequence for a positive integer $q \geq 2$. Then, $a(n)$ is a q th root of unity for any integer $0 \leq n < N$. The *second-order correlation measure* can be extended for polyphase sequences as follows.

Definition 1. For a polyphase sequence $\{a(n)\}$ of period N , the *second-order correlation measure* of $\{a(n)\}$ is defined as

$$C_2(a) = \max_{\substack{0 \leq i_1 < i_2 < N, \\ 1 \leq M \leq N - i_2}} \left| \sum_{n=0}^{M-1} a(n + i_1) a(n + i_2)^* \right|, \quad (1)$$

where i_1, i_2 and M are all integers, and $*$ denotes the complex conjugation.

As a measure of the pseudo-randomness of a single sequence $\{a(n)\}$, the *partial-period autocorrelation* $R_a(\tau, k; B)$ is defined as

$$R_a(\tau, k; B) = \sum_{n=k}^{k+B-1} a(n) a^*(n + \tau), \quad (2)$$

for an integer $0 \leq \tau < N$, an integer $0 \leq k < N$ and an integer $1 \leq B \leq N$ where all operations among the indices are computed modulo N . It is clear that the well-known *periodic autocorrelation* $\theta_a(\tau)$ and the *aperiodic autocorrelation* $A_a(\tau)$ can be expressed in terms of $R_a(\tau, k; B)$ as

$$\theta_a(\tau) = R_a(\tau, k; N) \text{ and } A_a(\tau) = R_a(\tau, 0; N - \tau)$$

for any $0 \leq \tau < N$ and any $0 \leq k < N$.

Setting $i_1 = k$, $i_2 - i_1 = \tau$ and $M = B$ in (1), we get

$$\begin{aligned}
C_2(a) &= \max_{0 \leq k < N} \max_{1 \leq \tau \leq N-1-k} \left| \sum_{n=0}^{B-1} a(n+k)a(n+k+\tau)^* \right| \\
&= \max_{1 \leq B \leq N-(k+\tau)} |R_a(\tau, k; B)|. \quad (3)
\end{aligned}$$

In other words, $C_2(a)$ can be viewed as the maximum magnitude of the (out-of-phase) partial-period autocorrelation of $\{a(n)\}$ with a limited range of τ and B depending on k . Since $C_2(a)$ can be expressed in terms of $R_a(k, \tau; B)$, we focus on the theory of partial-period autocorrelation as a first step for investigating $C_2(a)$.

Paterson and Lothian [9] gave an alternative expression of the partial-period correlation by using the window sequence. Recently, Lee and Yang [10] generalized their idea by introducing the so-called *induced correlation* and investigated the partial-period correlations of Zadoff-Chu sequences. For two integers k and B , where $0 \leq k \leq N-1$ and $0 < B \leq N$, define the window sequence $\{\lambda_{k,B}(n)\}$ as

$$\lambda_{k,B}(n) = \mathbf{I}_{A(k,B)}(n)$$

where $A(k, B)$ is the subset of $\mathbb{Z}_N$ defined by

$$A(k, B) = \begin{cases} \{n \mid k \leq n \leq k+B-1\}, & \text{if } k+B-1 < N \\ \{n \mid 0 \leq n \leq \langle k+B-1 \rangle_N \\ \text{or } k \leq n < N\}, & \text{if } k+B-1 \geq N \end{cases}$$

and $\mathbf{I}_S(\cdot)$ denotes the indicator function of a set S defined by

$$\mathbf{I}_S(n) = \begin{cases} 1, & \text{if } n \in S \\ 0, & \text{otherwise.} \end{cases}$$

Then the partial-period autocorrelation can be expressed as

$$R_a(\tau, k; B) = \frac{1}{N} \sum_{l=0}^{N-1} \hat{\lambda}_{k,B}(l) \theta_{a^l, a}(\tau). \quad (4)$$

Here, $\{\hat{\lambda}_{k,B}(l)\}_{l=0}^{N-1}$ is the discrete Fourier transform (DFT) sequence of $\{\lambda_{k,B}(n)\}_{n=0}^{N-1}$ defined by

$$\hat{\lambda}_{k,B}(l) = \sum_{n=0}^{N-1} \lambda_{k,B}(n) W_N[nl], \quad 0 \leq l < N,$$

and $\{\theta_{a^l, a}(\tau)\}_{l=0}^{N-1}$ is referred to as the l th induced correlation of $\{a(n)\}_{n=0}^{N-1}$ defined by

$$\theta_{a^l, a}(\tau) = \sum_{n=0}^{N-1} a(n) W_N[-nl] a^*(\langle n+\tau \rangle_N)$$

with $W_N[x] = \exp(2\pi j x \sqrt{-1}/N)$.

For a given k , $0 \leq k < N$, we define the following three $N \times N$ matrices $\mathbf{R}_a^{(k)}$, $\hat{\mathbf{A}}^{(k)}$ and Θ_a as

$$\begin{aligned}
\mathbf{R}_a^{(k)} &\triangleq \begin{bmatrix} R_a(0, k; 1) & R_a(1, k; 1) & \cdots & R_a(N-1, k; 1) \\ R_a(0, k; 2) & R_a(1, k; 2) & \cdots & R_a(N-1, k; 2) \\ \vdots & \vdots & \ddots & \vdots \\ R_a(0, k; N) & R_a(1, k; N) & \cdots & R_a(N-1, k; N) \end{bmatrix}, \\
\hat{\mathbf{A}}^{(k)} &\triangleq \begin{bmatrix} \hat{\lambda}_{k,1}(0) & \hat{\lambda}_{k,1}(1) & \cdots & \hat{\lambda}_{k,1}(N-1) \\ \hat{\lambda}_{k,2}(0) & \hat{\lambda}_{k,2}(1) & \cdots & \hat{\lambda}_{k,2}(N-1) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\lambda}_{k,N}(0) & \hat{\lambda}_{k,N}(1) & \cdots & \hat{\lambda}_{k,N}(N-1) \end{bmatrix}, \quad (5)
\end{aligned}$$

and

$$\Theta_a = \begin{bmatrix} \theta_{a^0, a}(0) & \theta_{a^0, a}(1) & \cdots & \theta_{a^0, a}(N-1) \\ \theta_{a^1, a}(0) & \theta_{a^1, a}(1) & \cdots & \theta_{a^1, a}(N-1) \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{a^{N-1}, a}(0) & \theta_{a^{N-1}, a}(1) & \cdots & \theta_{a^{N-1}, a}(N-1) \end{bmatrix},$$

respectively. Then the relation in (4) leads to

$$\mathbf{R}_a^{(k)} = \frac{1}{N} \hat{\mathbf{A}}^{(k)} \Theta_a. \quad (6)$$

The matrix representation in (6) gives us a much more intuitive way of understanding partial-period correlation and analyzing the second-order correlation measure, as will be shown in the next section.

3. Zadoff-Chu Sequences and Their Second-Order Correlation Measure

As seen in the previous section, the second-order correlation measure is closely related to the magnitude of partial-period autocorrelation. In this section, we will review Zadoff-Chu sequences and investigate their second-order correlation measure.

3.1 Partial-Period Autocorrelation of Zadoff-Chu Sequences

In this subsection, we briefly review the property of the partial-period autocorrelation of Zadoff-Chu sequences, which will be useful for deriving our main result.

Definition 2 ([11], [12]). For a positive integer N and an integer $1 \leq s < N$ with $\gcd(N, s) = 1$, the Zadoff-Chu sequence $\{c_s(n)\}$ of period N is defined as

$$c_s(n) = W_N \left[\frac{sn(n + \langle N \rangle_2)}{2} \right].$$

According to Definition 2, there exist $\phi(N)$ Zadoff-Chu sequences of period N where $\phi(N)$ is the Euler totient function of N defined as $\phi(N) = |\{1 \leq n < N \mid \gcd(n, N) = 1\}|$ [13]. It is known in [11] that Zadoff-Chu sequences of the same period have the same periodic autocorrelation regardless of s . On the other hand, their aperiodic autocorrelations depend on the choice of s . Regarding partial-period autocorrelation, $|\{R_{c_{s_1}}(\tau, k; B)\}| \neq |\{R_{c_{s_2}}(\tau, k; B)\}|$ in general if $s_1 \neq s_2$, but $\{c_{s_1}(n)\}$ and $\{c_{s_2}(n)\}$ have the same distribution of magnitudes of partial-period autocorrelation for any

$$|\hat{\mathbf{A}}_{-1}^{(0)}| = \begin{bmatrix} |\hat{\lambda}_{0,1}(1)| & |\hat{\lambda}_{0,1}(2)| & |\hat{\lambda}_{0,1}(3)| & |\hat{\lambda}_{0,1}(4)| & |\hat{\lambda}_{0,1}(5)| & |\hat{\lambda}_{0,1}(6)| \\ |\hat{\lambda}_{0,2}(1)| & |\hat{\lambda}_{0,2}(2)| & |\hat{\lambda}_{0,2}(3)| & |\hat{\lambda}_{0,2}(4)| & |\hat{\lambda}_{0,2}(5)| & |\hat{\lambda}_{0,1}(6)| \\ |\hat{\lambda}_{0,3}(1)| & |\hat{\lambda}_{0,3}(2)| & |\hat{\lambda}_{0,3}(3)| & |\hat{\lambda}_{0,3}(4)| & |\hat{\lambda}_{0,3}(5)| & |\hat{\lambda}_{0,1}(6)| \\ |\hat{\lambda}_{0,4}(1)| & |\hat{\lambda}_{0,4}(2)| & |\hat{\lambda}_{0,4}(3)| & |\hat{\lambda}_{0,4}(4)| & |\hat{\lambda}_{0,4}(5)| & |\hat{\lambda}_{0,1}(6)| \\ |\hat{\lambda}_{0,5}(1)| & |\hat{\lambda}_{0,5}(2)| & |\hat{\lambda}_{0,5}(3)| & |\hat{\lambda}_{0,5}(4)| & |\hat{\lambda}_{0,5}(5)| & |\hat{\lambda}_{0,5}(6)| \\ |\hat{\lambda}_{0,6}(1)| & |\hat{\lambda}_{0,6}(2)| & |\hat{\lambda}_{0,6}(3)| & |\hat{\lambda}_{0,6}(4)| & |\hat{\lambda}_{0,6}(5)| & |\hat{\lambda}_{0,6}(6)| \end{bmatrix}. \quad (7)$$

k and B [10]. Throughout the paper, we first consider the case of $s = 1$ and denote $\{c_1(n)\}$ simply by $\{c(n)\}$. It will turn out later that its second-order correlation measure does not depend on the value of s .

For an $M \times N$ matrix $\mathbf{A} \triangleq (a_{ij})$ over the set of complex numbers, $|\mathbf{A}|$ and $\mathbf{A}^T$ stand for its magnitude matrix given by $|\mathbf{A}| = (|a_{ij}|)$ and its transpose, respectively. The following lemma gives important information on $|\{R_c(\tau, k; B)\}|$ of the matrix form.

Lemma 3. Let $\{c(n)\}$ be the Zadoff-Chu sequence of period N with $s = 1$. Then

$$|\mathbf{R}_c^{(k)}| = |\hat{\mathbf{A}}^{(0)}| \quad (8)$$

for any $0 \leq k < N$.

Proof. It follows from Theorem 4 and Lemma 5 in [10] that

$$|\Theta_c| = N\mathbf{P}_\sigma^T \quad (9)$$

where $\mathbf{P}_\sigma$ is the $N \times N$ permutation matrix associated with the permutation σ on $\mathbb{Z}_N$ defined by

$$\begin{aligned} \sigma(x) &= \langle -sx \rangle_N \\ &= \langle -x \rangle_N, \quad x \in \mathbb{Z}_N, \end{aligned} \quad (10)$$

i.e., $\mathbf{P}_\sigma^T$ is of the form

$$\mathbf{P}_\sigma^T = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \cdots & 0 & 0 \end{bmatrix}.$$

The matrix $|\hat{\mathbf{A}}^{(k)}|$ in (5) is decomposed as

$$\begin{aligned} |\hat{\mathbf{A}}^{(k)}| &= [|\hat{\lambda}_0^{(k)}|, |\hat{\lambda}_1^{(k)}|, \dots, |\hat{\lambda}_{N-1}^{(k)}|] \\ &= [|\hat{\lambda}_0^{(0)}|, |\hat{\lambda}_1^{(0)}|, \dots, |\hat{\lambda}_{N-1}^{(0)}|] \end{aligned} \quad (11)$$

where $|\hat{\lambda}_l^{(k)}|$ denotes the l th column of $|\hat{\mathbf{A}}^{(k)}|$. The equality in (11) comes from the fact that each entry of $|\hat{\mathbf{A}}^{(k)}|$ is equal to that of $|\hat{\mathbf{A}}^{(0)}|$ for all k . Moreover, Lemma 11 in [10] tells that

$$|\hat{\lambda}_l^{(0)}| = |\hat{\lambda}_{N-l}^{(0)}| \quad (12)$$

for $1 \leq l < N$. By combining (6), (9) and (12), we get the desired result. $\square$

3.2 Second-Order Correlation Measure of Zadoff-Chu Sequences

Based on the relation in (3) and Lemma 3, we are now ready to compute the second-order correlation measure of Zadoff-Chu sequences.

Theorem 4. Let $\{c(n)\}$ be the Zadoff-Chu sequence of period N with $s = 1$. Then its second-order correlation measure is given by

$$C_2(c) = \begin{cases} \frac{1}{\sin(\frac{\pi}{N})}, & \text{if } \langle N \rangle_2 = 0, \\ \frac{|\sin(\frac{\pi}{N} \frac{N \pm 1}{2})|}{\sin(\frac{\pi}{N})}, & \text{if } \langle N \rangle_2 = 1. \end{cases}$$

Proof. From (3) and (8), we have

$$\begin{aligned} C_2(a) &= \max_{0 \leq k < N} \max_{1 \leq \tau \leq N-1-k} \max_{1 \leq B \leq N-(k+\tau)} |R_a(\tau, k; B)| \\ &= \max_{0 \leq k < N} \max_{1 \leq \tau \leq N-1-k} \max_{1 \leq B \leq N-(k+\tau)} |\hat{\lambda}_{0,B}(\tau)| \\ &= \max_{1 \leq \tau \leq N-1} \max_{1 \leq B \leq N-\tau} |\hat{\lambda}_{0,B}(\tau)|. \end{aligned} \quad (13)$$

For an easier presentation, consider the $(N-1) \times (N-1)$ matrix $|\hat{\mathbf{A}}_{-1}^{(0)}|$ which is obtained by deleting the first column and the last row from $|\hat{\mathbf{A}}^{(0)}|$ in (5). We then easily see that $C_2(c)$ in (13) is the maximum component among the upper-left part of $|\hat{\mathbf{A}}_{-1}^{(0)}|$. Furthermore, it is shown in [10] that the maximum component in each row of $|\hat{\mathbf{A}}_{-1}^{(0)}|$, i.e., $[|\hat{\lambda}_{0,B}(1)|, |\hat{\lambda}_{0,B}(2)|, \dots, |\hat{\lambda}_{0,B}(N-1)|]$, is $|\hat{\lambda}_{0,B}(1)|$. Hence, we get

$$\begin{aligned} C_2(c) &= \max_{1 \leq B \leq N-1} |\hat{\lambda}_{0,B}(1)| \\ &= \max_{1 \leq B \leq N-1} \frac{|\sin(\frac{\pi}{N} B)|}{\sin(\frac{\pi}{N})} \\ &= \begin{cases} \frac{1}{\sin(\frac{\pi}{N})}, & \text{if } \langle N \rangle_2 = 0, \\ \frac{|\sin(\frac{\pi}{N} \frac{N \pm 1}{2})|}{\sin(\frac{\pi}{N})}, & \text{if } \langle N \rangle_2 = 1. \end{cases} \end{aligned}$$

where the second equality comes from the fact that $|\hat{\lambda}_{0,B}(\tau)| = \frac{|\sin(\frac{\pi}{N} B \tau)|}{\sin(\frac{\pi}{N} \tau)}$. $\square$

Example 5. Let $\{c(n)\}_{n=0}^6$ be the Zadoff-Chu sequence of period $N = 7$ with $s = 1$. As a heuristic approach to $C_2(c)$, consider the matrix $|\hat{\mathbf{A}}_{-1}^{(0)}|$ discussed in Proof of Theorem 4

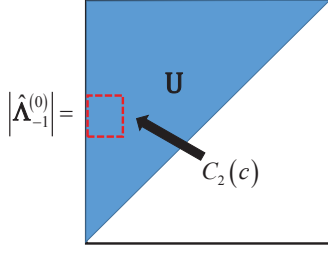


Fig. 1 Relation between $C_2(c)$ and $|\hat{\Lambda}_{-1}^{(0)}|$.

as in (7). Let $\mathbf{U}$ be the upper-left part of $|\hat{\Lambda}_{-1}^{(0)}|$. As mentioned, $C_2(c)$ is the maximum entry of $\mathbf{U}$ which is ‘always in the middle’ on the first column of $|\hat{\Lambda}_{-1}^{(0)}|$. In this case,

$$\begin{aligned} C_2(c) &= |\hat{\lambda}_{0,3}(1)| \\ &= |\hat{\lambda}_{0,4}(1)| \\ &= \frac{|\sin(\frac{4\pi}{7})|}{\sin(\frac{\pi}{7})} \\ &= 2.2470. \end{aligned}$$

Figure 1 illustrates the relation between $C_2(c)$ and $|\hat{\Lambda}_{-1}^{(0)}|$. ■

A natural question is to ask whether the second-order correlation measure of Zadoff-Chu sequences depend on the choice of s or not. The following theorem gives an answer that $C_2(c_s)$ is independent of the choice of s .

Theorem 6. Let $\{c_s(n)\}$ be the Zadoff-Chu sequence of period N for a positive integer s such that $\gcd(N, s) = 1$. Then its second-order correlation measure is given by

$$\begin{aligned} C_2(c_s) &= C_2(c_1) \\ &= \begin{cases} \frac{1}{\sin(\frac{\pi}{N})}, & \text{if } \langle N \rangle_2 = 0, \\ \frac{|\sin(\frac{\pi}{N} \frac{N \pm 1}{2})|}{\sin(\frac{\pi}{N})}, & \text{if } \langle N \rangle_2 = 1. \end{cases} \end{aligned}$$

Proof. We only prove the case that $\langle N \rangle_2 = 0$ since the other case can be similarly proved. From (6), (9) and (10), we get

$$\begin{aligned} |\mathbf{R}_{c_s}^{(k)}| &= |\hat{\Lambda}^{(0)} \mathbf{P}_{\sigma}^T| \\ &= |\hat{\Lambda}^{(0)} \mathbf{P}_{\sigma^{-1}}| \end{aligned} \quad (14)$$

where $\mathbf{P}_{\sigma^{-1}}$ is the $N \times N$ permutation matrix associated with the permutation σ^{-1} on $\mathbb{Z}_N$ defined by

$$\sigma^{-1}(x) = \langle -s^{-1}x \rangle_N, \quad x \in \mathbb{Z}_N.$$

Let $\sigma^{-1} \triangleq \sigma'$ and $s^{-1} \bmod N \triangleq s'$. Then (14) becomes

$$|\mathbf{R}_{c_s}^{(k)}| = |\hat{\Lambda}^{(0)} \mathbf{P}_{\sigma'}|.$$

Similarly to the case that $s = 1$, we consider the $(N-1) \times (N-1)$ matrix $|\mathbf{R}_{c_s, -1}^{(k)}|$ which is obtained by deleting the first column and the last row from $|\mathbf{R}_{c_s}^{(k)}|$. Note that $C_2(c_s)$

is also the maximum component among the upper-left part of $|\mathbf{R}_{c_s, -1}^{(k)}|$. From (12), we have

$$\begin{aligned} \max_{1 \leq \tau, B \leq N-1} |\hat{\lambda}_{0,B}(\tau)| &= |\hat{\lambda}_{0,N/2}(1)| \\ &= |\hat{\lambda}_{0,N/2}(N-1)|. \end{aligned}$$

Therefore, it suffices to show that there is no permutation σ' on $\mathbb{Z}_N$ such that

$$\sigma'(1) > \frac{N}{2} \quad \text{and} \quad \sigma'(N-1) > \frac{N}{2}. \quad (15)$$

Suppose that a permutation σ' satisfies (15). From $\sigma'(1) > N/2$, we get

$$\langle -s' \rangle_N > N/2 \Leftrightarrow N/2 > s'. \quad (16)$$

At the same time, from $\sigma'(N-1) > N/2$, we have

$$\langle -s'(N-1) \rangle_N > N/2 \Leftrightarrow s' > N/2$$

which contradicts (16), so we get the desired result. □

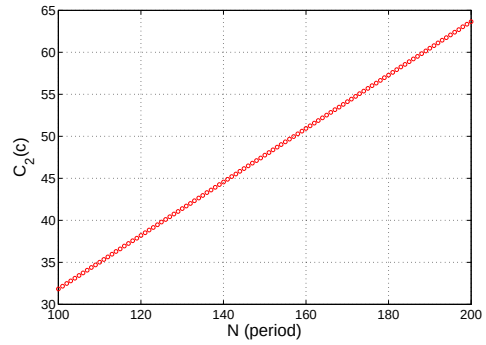


Fig. 2 Second-order correlation measure $C_2(c)$ of Zadoff-Chu sequences of period N , $100 \leq N \leq 200$.

By Theorems 4 and 6, we see that the second-order correlation measure of Zadoff-Chu sequences does not depend on the choice of s , unlike the aperiodic autocorrelation. Figure 2 shows the value of $C_2(c)$ for the period between 100 and 200. Since Zadoff-Chu sequences are a family of polyphase sequences with perfect periodic autocorrelation, their second-order correlation measure may act as an upper bound on that of a best polyphase sequence with perfect periodic autocorrelation.

4. Conclusions

Motivated by the r th-order correlation measure for cryptography, we linked the second-order correlation measure to the partial-period autocorrelation. We then investigated the second-order correlation measure of a Zadoff-Chu sequence $\{c_s(n)\}$. As a result, we evaluated the exact value of $C_2(c_s)$ for any period N . In addition, we confirmed the invariance of $C_2(c_s)$ to the choice of s for a given period N . Future research may include the application of $C_2(c_s)$ and the determination of the r th-order correlation measure for $r \geq 3$.

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